# James Foehner

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## EDUCATION

**Northeastern University, Khoury College of Computer Sciences** *Boston, MA*
*Candidate for BS in Data Science* *Expected May 2027*
***Coursework:*** Programming with Data Science, Machine Learning & Data Mining, Programming in C++, Statistics for Economists, Algorithms, Advanced Machine Learning (in progress)

## SKILLS

**Languages:** Python | R | C++
**Technical:** Supervised & Reinforcement Learning | Predictive Modeling | LLM Fine-Tuning & Evaluation | Data Visualization | SQL
**Finance:** Financial Modeling (Excel) | Valuation (Comps, Sum-of-the-Parts) | Event Studies | WRDS (CRSP, Compustat) | SEC EDGAR

## PERSONAL PROJECTS

**Activist Hedge Fund Target Screener & Post Holdings Activist Pitch** *Oct 2026 - Nov 2026*
*Python, SQL, WRDS (CRSP, Compustat), SEC EDGAR, Excel, Claude API | github.com/jamefoehner/activist-target-screener*

- Built a Python pipeline collecting 232 activist campaigns (2010–26) from SEC 13D filings; an event study on CRSP returns found 7.0% abnormal returns around announcements (t = 13.3), with one-year gains concentrated in targets later acquired (+19% vs. ~0%)
- Trained a gradient-boosting model on 44K firm-years of peer-relative Compustat data with walk-forward validation; top 10% of predictions caught 27% of future targets (AUC 0.73), ranking E2open in the top 0.4% before Elliott's 2023 campaign
- Used the model to identify Post Holdings (ranked 7th of 2,600+) and built an Excel model comparing its debt-funded buybacks with a plan to sell Weetabix and delever (~$111/share vs. $77; stronger in a downside case)
- Tested two thesis assumptions against data (a deleveraging re-rating; spin-off value across 182 spin-offs since 2010), found neither held, and rebuilt the pitch around capital allocation in an investment memo
- Built an LLM evaluation benchmark on 100 hand-labeled 13Ds: Claude Sonnet with a labeling guide fully matched reference labels on 83% of filings (20 of 20 vs. 18 of 20 for a second human); model disagreement found a 7% label error rate

**Activist Filing Monitor (live web app)** *Nov 2026*
*Python, Streamlit, Claude API, SEC EDGAR | activist-monitor.streamlit.app*

- Built and deployed a web app that pulls new SEC 13D filings from major activist investors, extracts each filing's stated purpose, and labels the demands with the Claude API; pipeline covered by 23 unit tests
- Joined the labels to a gradient-boosting activist-target screener so any of 2,600 companies can be looked up for its score, past campaigns and latest filings

**Reward Hacking in Reinforcement Learning for 13D Filing Analysis** *Nov 2026*
*Python, PyTorch, Hugging Face, LoRA, GRPO | github.com/jamefoehner/activist-target-screener*

- Fine-tuned Qwen2.5-1.5B with LoRA on Claude-labeled 13D filings to tag ten activist demands; macro-F1 on 89 hand-labeled filings rose from 0.35 to 0.67
- Wrote a GRPO training loop and showed reward hacking: RL from the base model collapsed under two naive rewards, answering "no demands" on 66 of 89 filings while scoring 87% accuracy
- Used paired bootstrap tests to show that starting RL from the fine-tuned model avoided collapse, but its best gain (0.73 macro-F1) was not distinguishable from fine-tuning alone

**NBA Prediction AI** *Jan 2026 - Apr 2026*

- Trained a Gradient Boosting Classifier for in-game win probability, achieving ROC-AUC 0.84 and Brier score 0.071, outperforming a Logistic Regression baseline by +0.08 AUC
- Engineered rolling-window features for three Random Forest regressors predicting player points, assists, and rebounds (±4.1-point MAE, 80% intervals from tree variance)
- Deployed an interactive 3-tab Streamlit app with real-time Plotly visualizations for live model inference and sensitivity analysis

**Aircraft Incident Database & Analytics** *May 2025 - Jul 2025*

- Designed a 3NF relational database for aircraft incident data, including the ERD, primary/foreign keys, and constraints
- Built an ETL pipeline in R to load and transform CSV data into a cloud-hosted MySQL database using batched SQL inserts and transaction management.
- Validated database integrity by comparing source CSV data against database records using counts, aggregates, and consistency tests